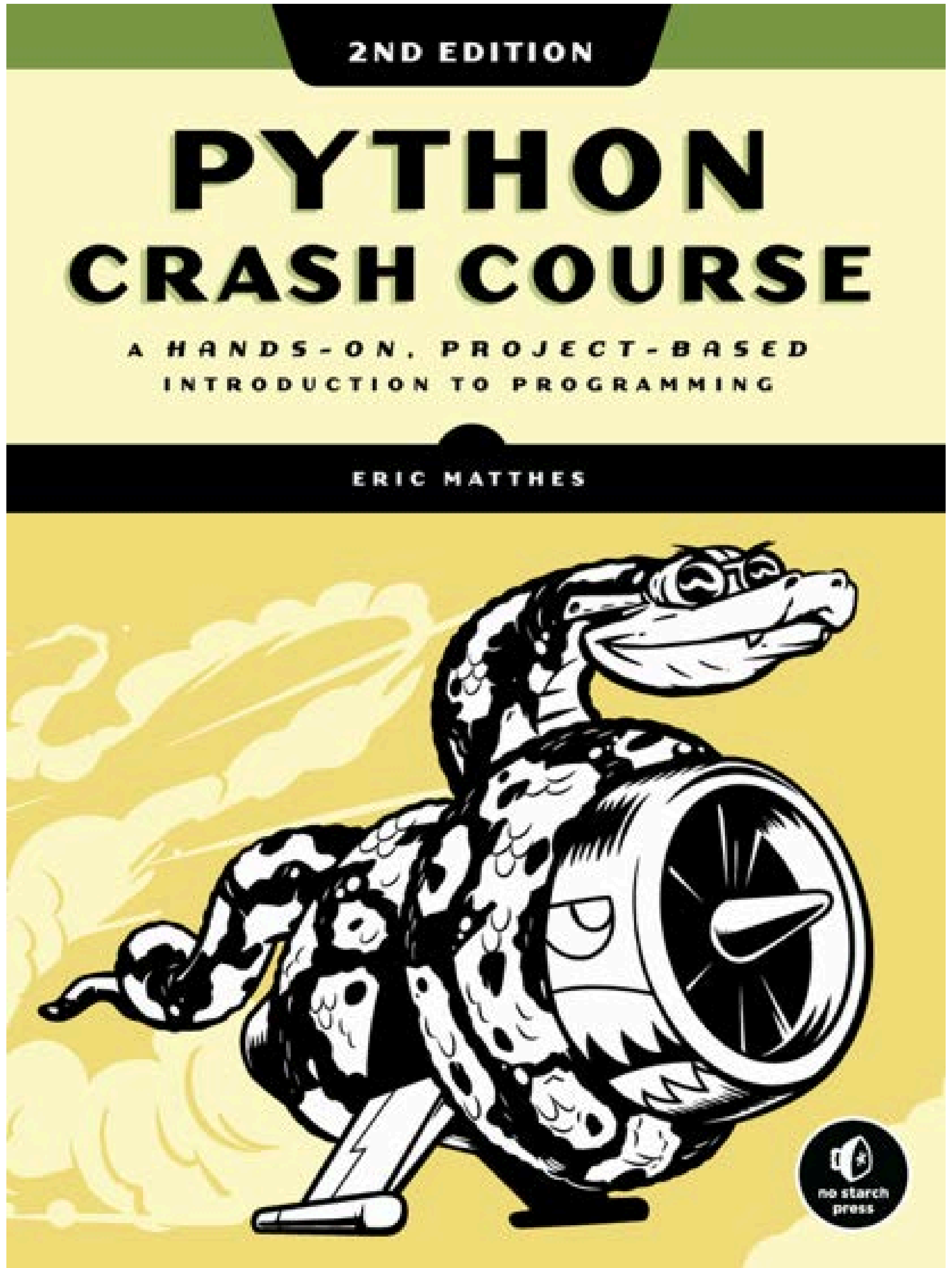
The following contents is based on these 2 main sources:



PYTHON BASIC

Introduction to python, structured in 10-week begineer programming program, design to understand the foundamental programming by python language.

BEFORE PYTHON

1

- Introduction to Computer things
- Installation set-up
- How to be ready with yr first python

START TO PYTHON

2

- Understand programming by python syntax
- Practices

ELEMENTARY IN PYTHON

3

- Introduction to logical things by caculating with condition
- Repetative concept with the given condition

MORE THAN ELEMENTARY

4

- Resuable concept in coding by python
- Compare & practice different types of function in python

FINALLYYYY

5

- A final project, required to apply things you have learnt

Things i have done with Python.

01

Data Analyse Role: Soil Nutrient Analysis

- Table dataset
- To explore and clean the large amount data, then prepare for prediction
- Build Machine Learning Model

02

Data Analyse Role: Amazon Production Review Analysis

- Raw text dataset from social platform
- To explore and clean with large amount to understant the behavior then apply prediction with Machine Learning

03

Automation with System Design: Auto Extract & Clean Table Data from pdf file to Machine Readable Format

- Backend: Design pipeline for processing
- Frontend: Simple UI for user to test

04

Recommendation Engine: MCT at Bamnang

- Design backend pipeline for frontend developer

05

Chatbot: BamPi Chatbot at Bamnang

- Start-to-end on backend pipeline for frontend developer
- Tutor Chatbot for student to do some practicals

There always candieees for
those work hard!

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Week1 - Basic Things before Hand on code

- Basic things about PC
- How PC communicate to PC?
- How PC understand Human?
- Why Programming?
- Why Python Language?

BASIC COMPUTER

To operate a computer, it require to have **Hardware** and **Software**.

- Hardware (CPU, RAM, Keyboard, motherboard, Monitor screen, mouse, trackpad, etc.) is the body.
- Software include the OS is the brain that tells everything how to work together.
    - E.g. Microsoft Windows 7 / 8 / 10, Mac OS Ventura 13.7
    - The applications are tools you use — and the OS helps them run.
    - E.g. Google Chrome, Microsoft Word, Telegram

HOW PC UNDERSTAND WE HAVE DONE ON THOSE HARDWARE?

How about Human talk to Human?

Human1: Cambodian
Language: Khmer & English

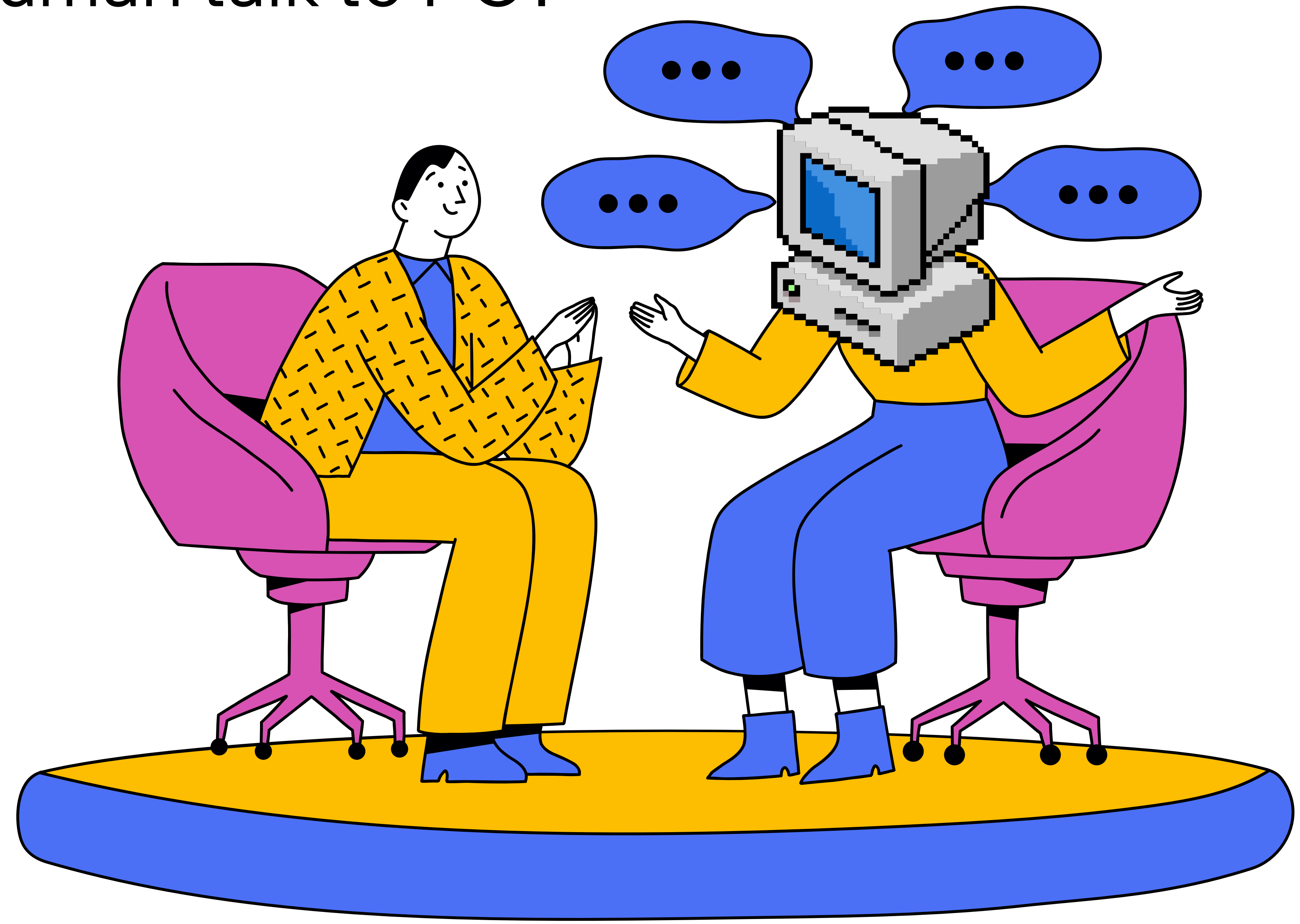
Human2: French
Language: French & English

HOW PC UNDERSTAND WE HAVE DONE ON THOSE HARDWARE?

How about Human talk to PC?

Humans think in:

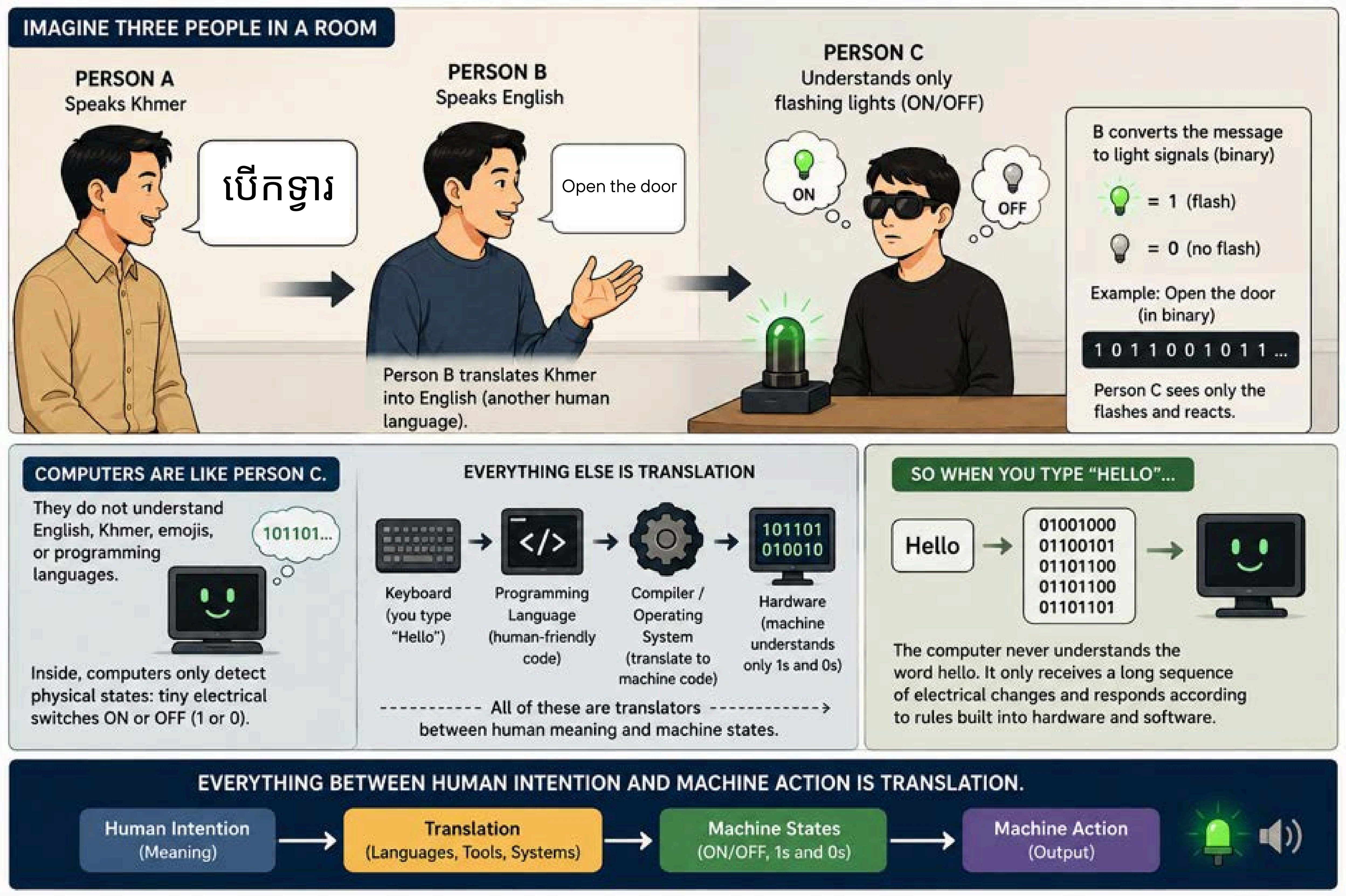
- ideas
- goals
- emotions
- meaning
- context
- ...



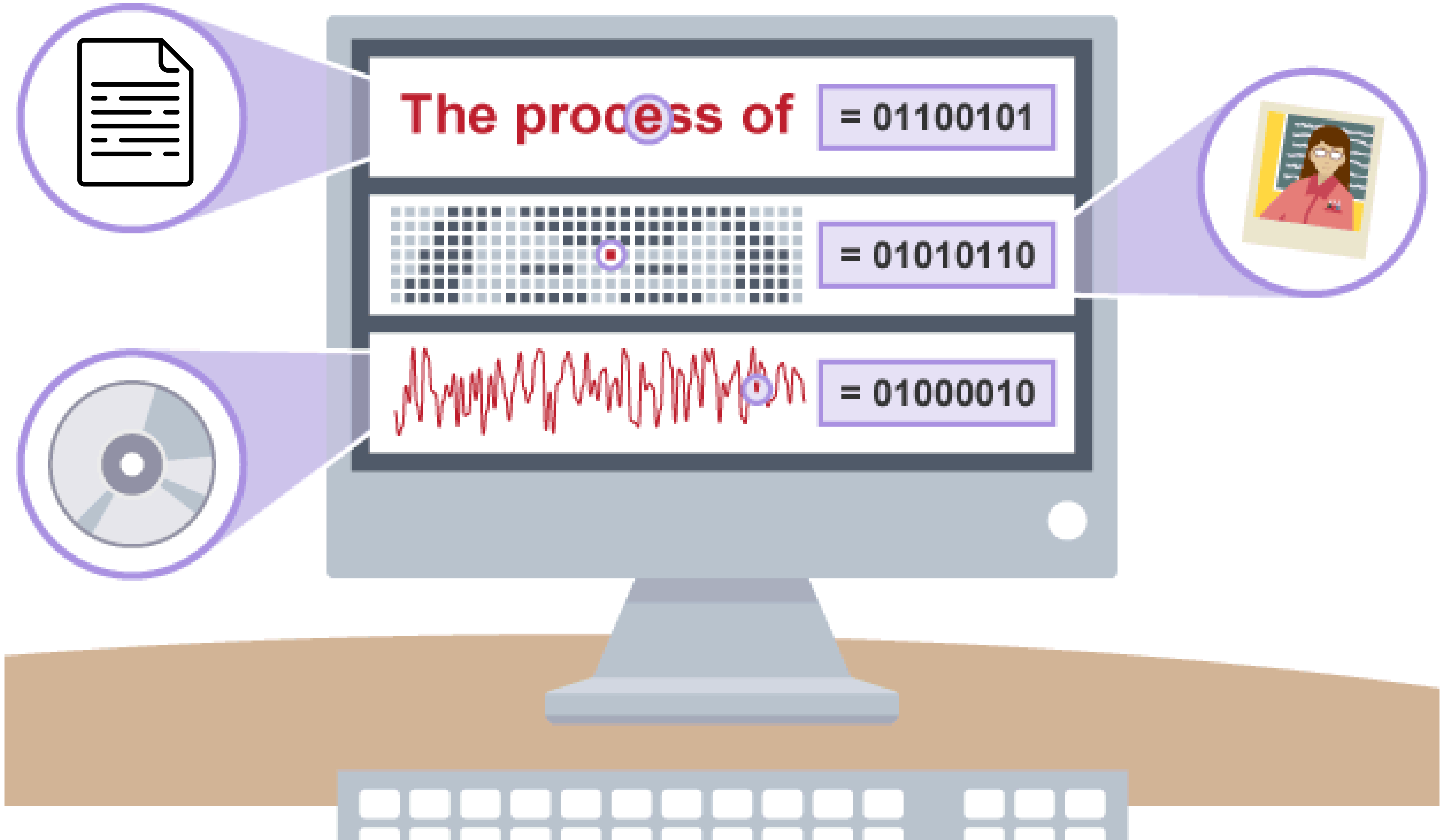
Computers operate on:

- electrical signals
- rules
- mathematics
- ...

How does meaning inside a human mind become electrical activity inside a machine, and then come back as something humans understand?



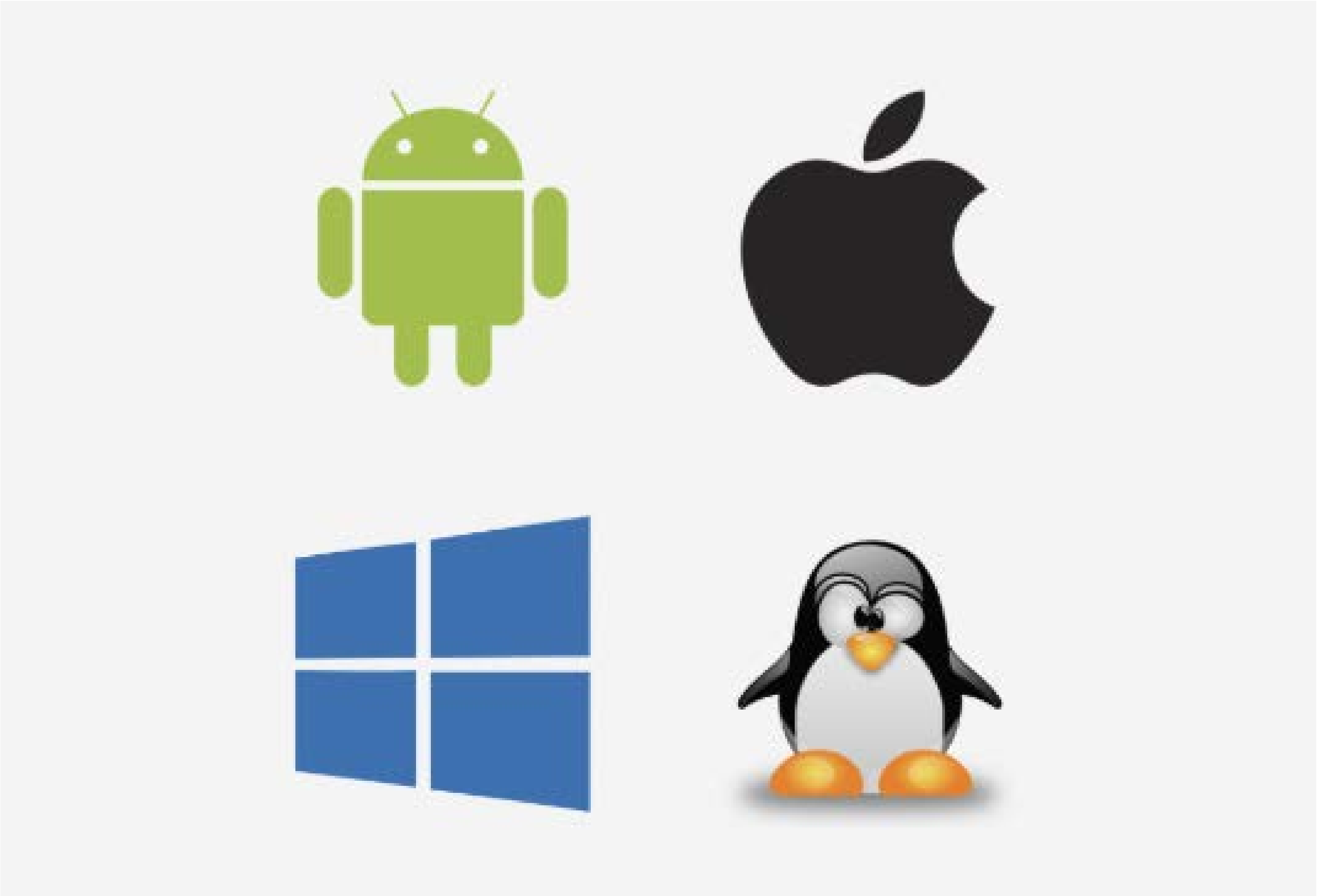
HOW PC UNDERSTAND WE HAVE DONE ON THOSE HARDWARE?



BASIC COMPUTER

What is an Operating System?

- An Operating System is the main software that manages all the hardware and software resources on a computer or device. It acts like a bridge between you (the user) and the computer's hardware.
- There are different operating systems created by different IT companies.



HOW PC INTERACT WITH THE OTHER PC???

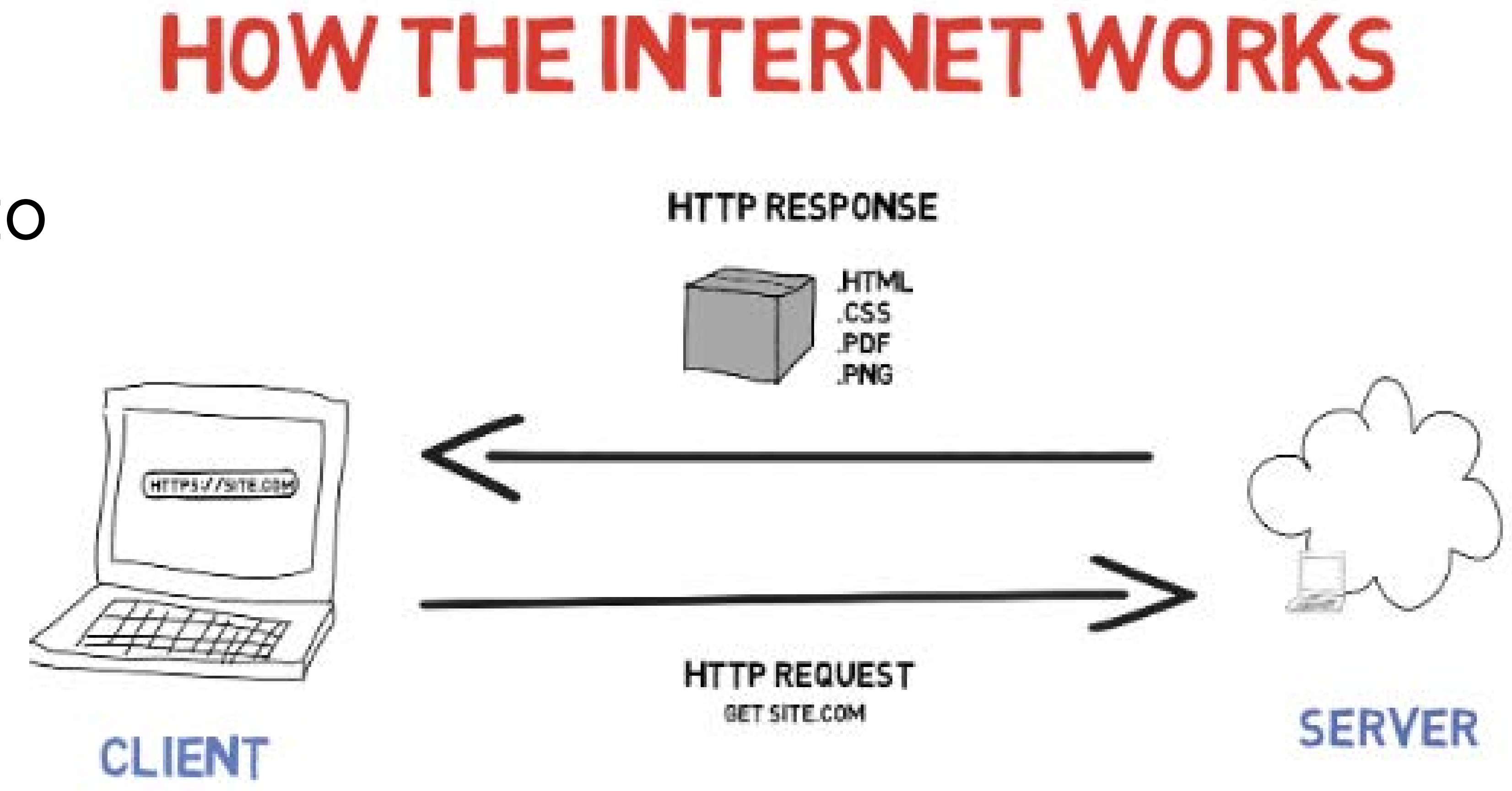
It just things, created from steel, how HUMAN manages things to let it communicate with one to another???

How Do We Access Websites?

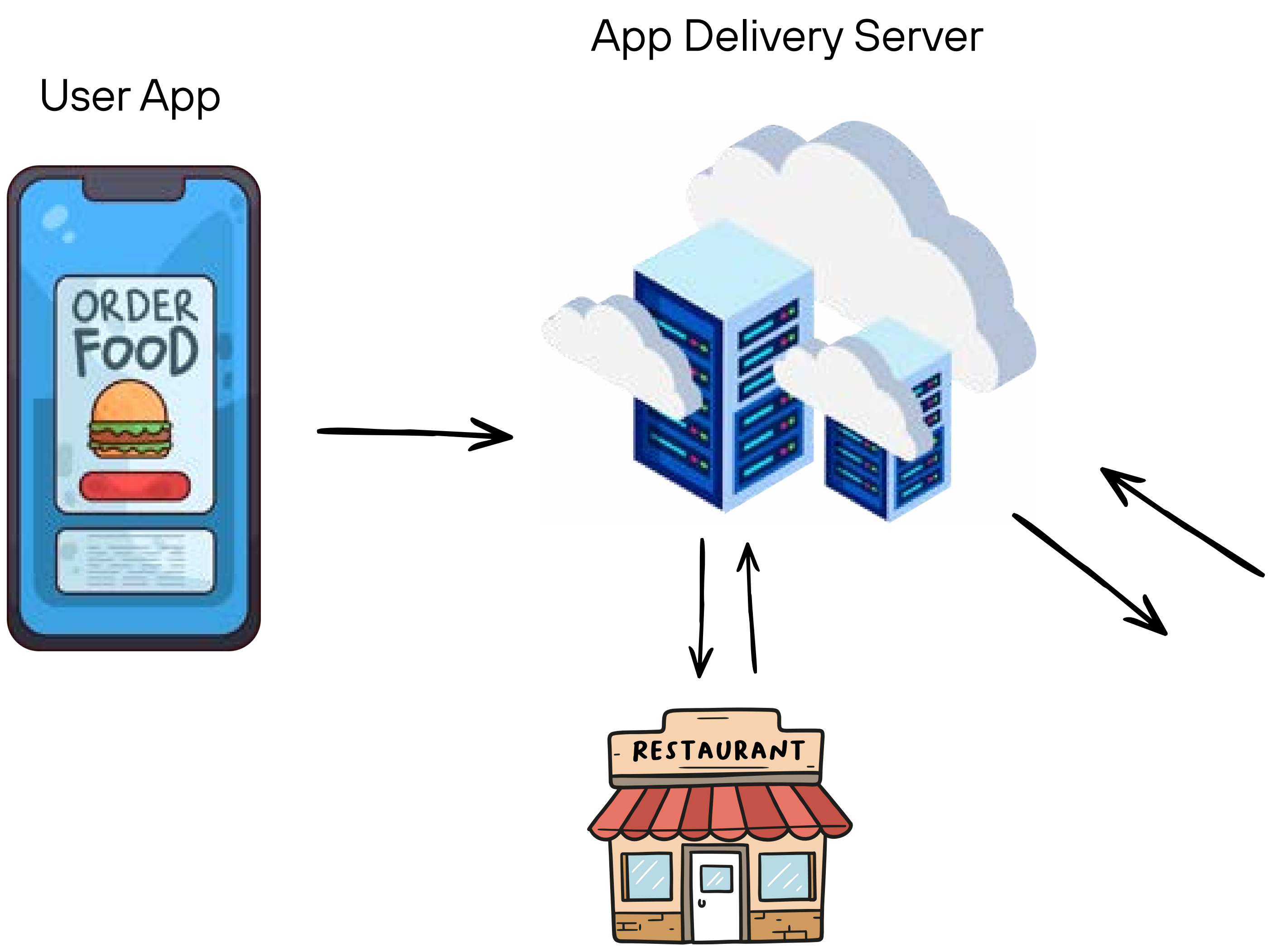
- Client (Your Computer/Phone) sends a request to a web server.
- Server processes the request and sends back a response.
- Browser renders the webpage for the user.

Key Terms:

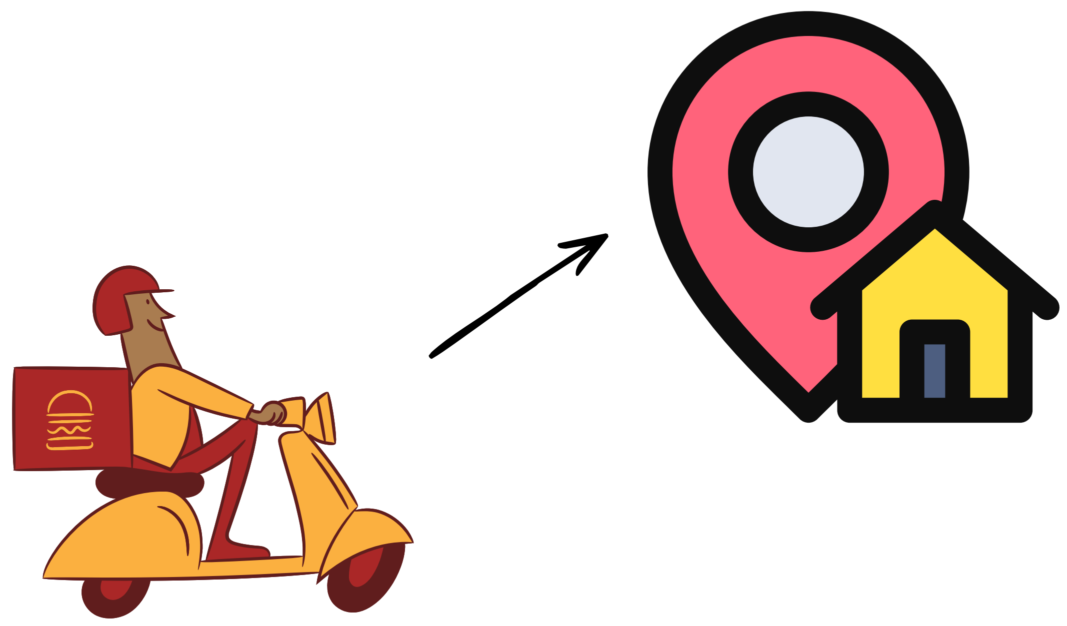
- Request: Asking for a webpage.
- Response: Receiving the requested data.
- IP Address: A unique number for each device on the internet.



HOW PC INTERACT WITH THE OTHER PC???

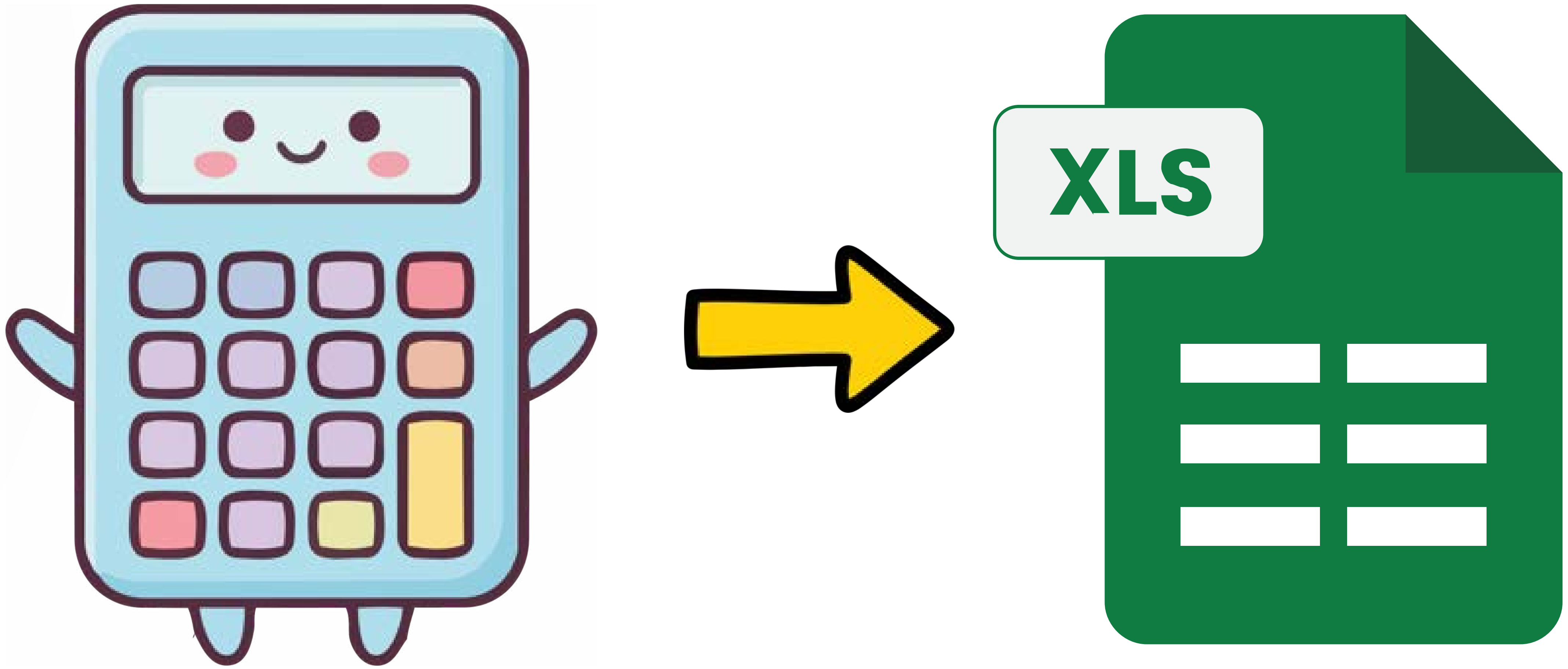


- Request = You ordering food through your phone through any Delivery App
- Server = Backend Delivery App send your REQUEST to Restaurant what you have ordered
- IP Address = Your home address for delivery
- Response = Food delivered to you successfully, then deliveryman call you to pick-up



WHY ??

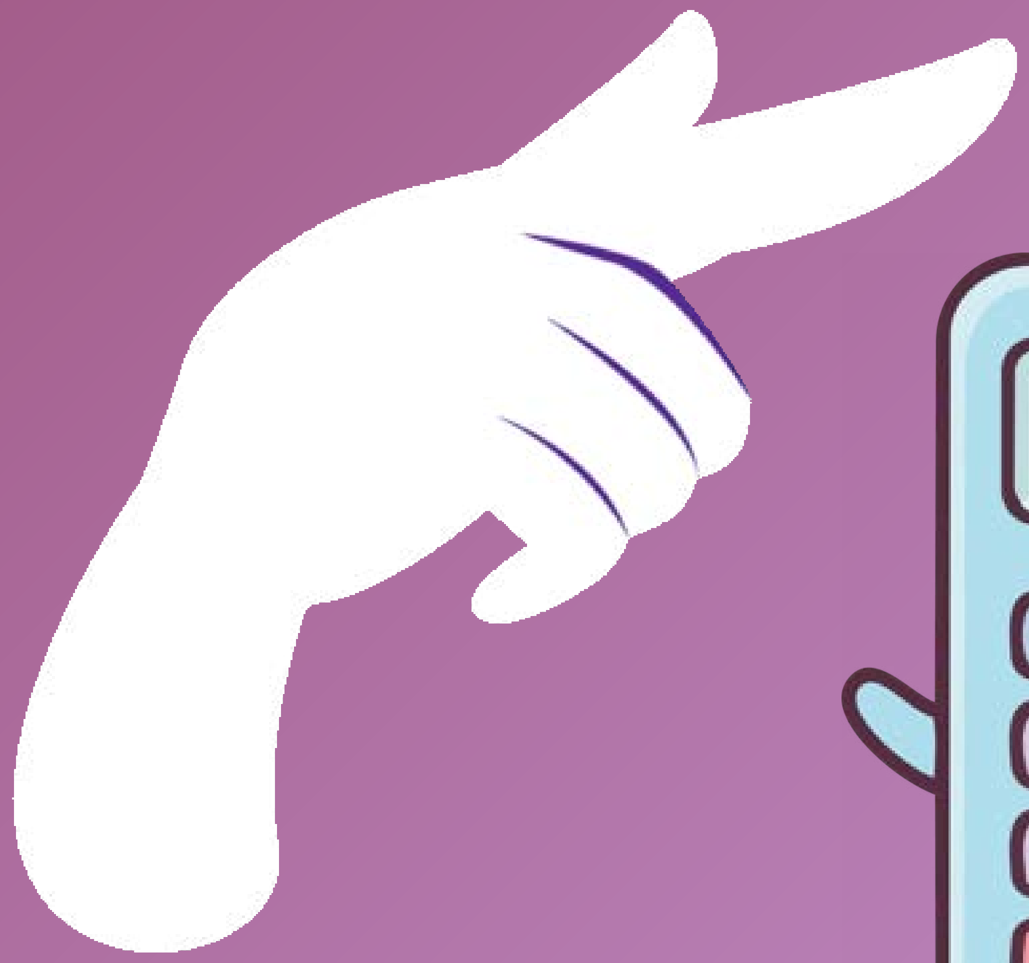
First... why programming?



WHITOUT PROGRAMMING

So, short without programming:

Human → Click → Compute → Repeat → Repeat → Repeat



Math Score	Code Score	Total
2	1	
3	2	
4	3	
5	4	
6	5	

WHY ??

Then... why Python??

Apply reverse engineer to **ask this WHY**?? What things that HUMAN need python noww?

Easy to Learn with simple syntax

```
print("Hello")
```

else programming language, like Java

```java
public class Main {
  public static void main(String[] args) {
    System.out.println("Hello");
  }
}
```

Build Things quickly with large communities developer support

- Games
- Websites
- AI tools
- Data projects

Powerful with automation tasks

- AI & Machine Learning
- Data Science
- Software Development
- Robotics
- Business and research

Week2 - Data Type

- Installation
- Data-Type
- Variable

INSTALLATION

Visual Studio Code

python™

ANACONDA®

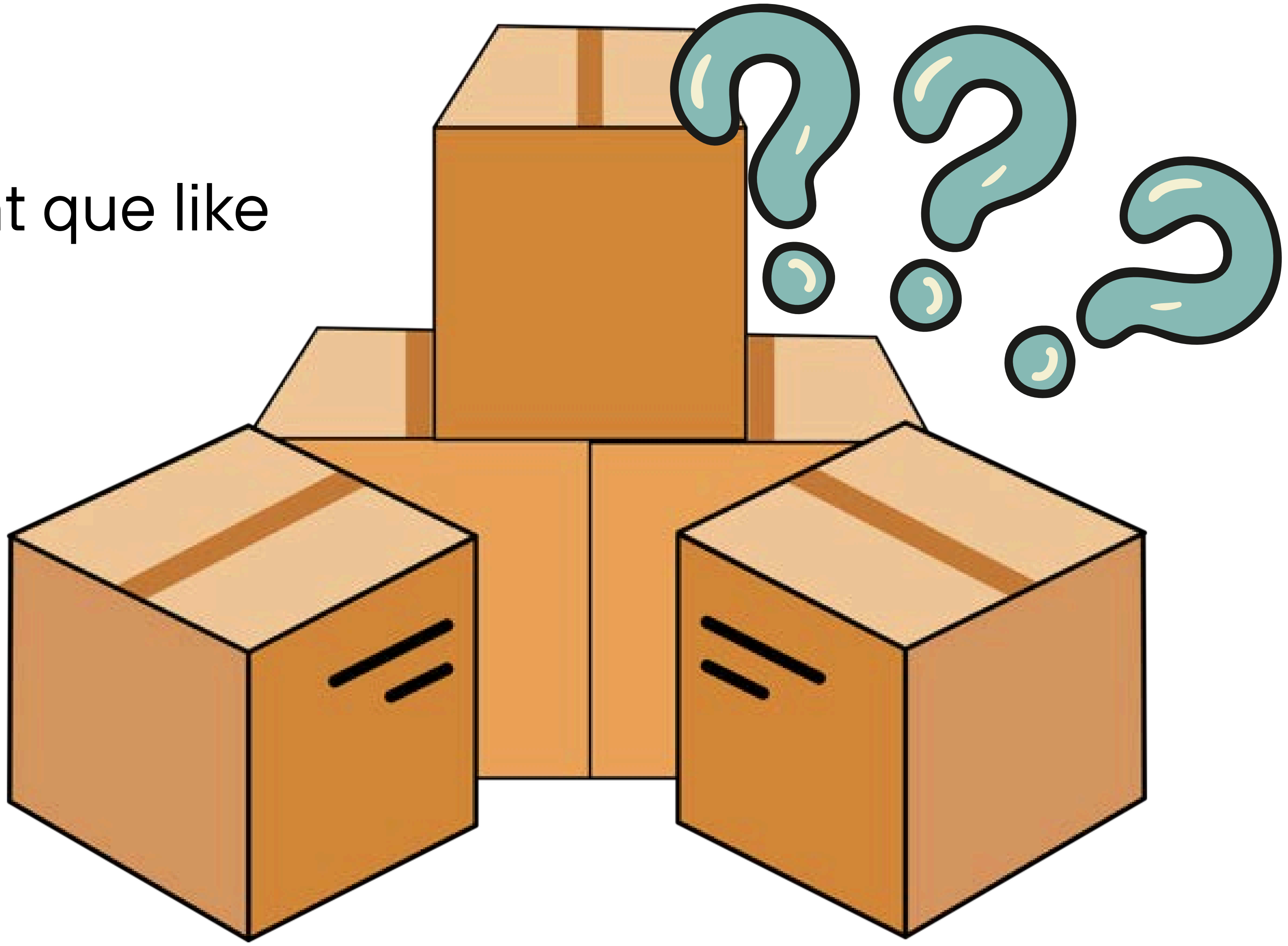
Discussion

Imagine you creating Facebook, Telegram, Linkedin or else application…

- First que come to developer is what kinds of information must the computer store?
- Explain

WHY DATA-TYPE?

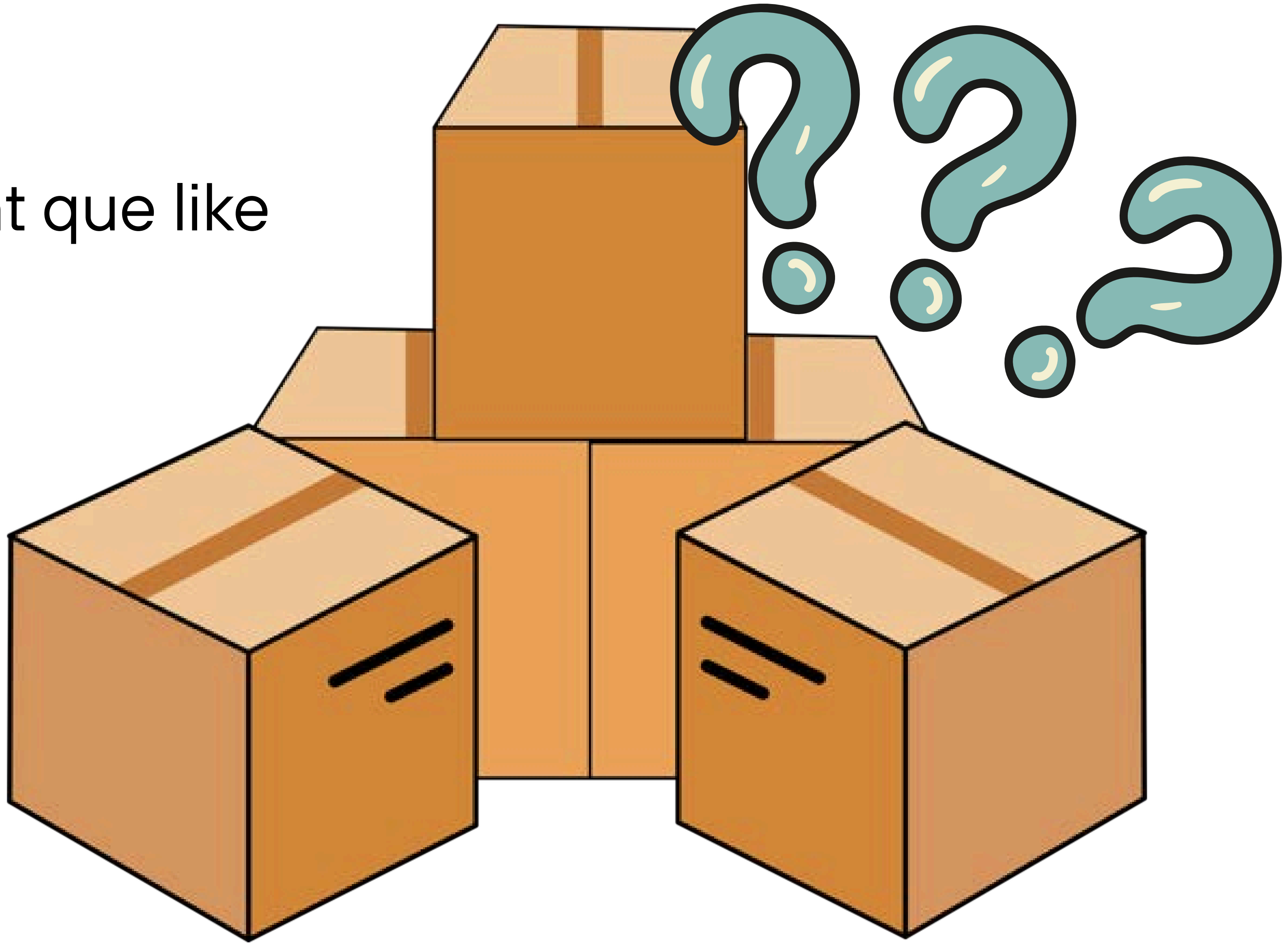
Imagine receiving a box without a label, you might que like what inside?



WHY DATA-TYPE?

Imagine receiving a box without a label, you might que like what inside?

so, the label tells you how to handle the box, and ***Data-Types are exactly that label***.



```
Python Data

├── Numeric
│       ├── int
│       └── float
│
├── Text
│       └── str
│
└── Logical
        └── bool
```

int	`age = 20`
float	`gpa = 3.85`
str	`name = "Name1"`
bool	`is_student = True`

WHAT IS VARIABLE??

Take a look at this math:

- **X** is a character, we declare to find its value

$$X + 5 = 10$$

In programming, variable is being used like **X**, we declare cuz we want to do and fine sth from it

HOW TO NAMING ON VARIABLES

- Variable names can contain only letters, numbers, and underscores
  - Ex: ***message_1*** but not ***1_message***
- Space is not allowed in variable name
  - Ex: ***name_variable*** not ***name variable***
- Avoid using python keyword
- Variable name should be short but descriptive
  - ***student_name*** is better than ***s_n***
- Better not to use **upper-case letter** (cuz naming as upper-case, it means sth else)

Week3 - List

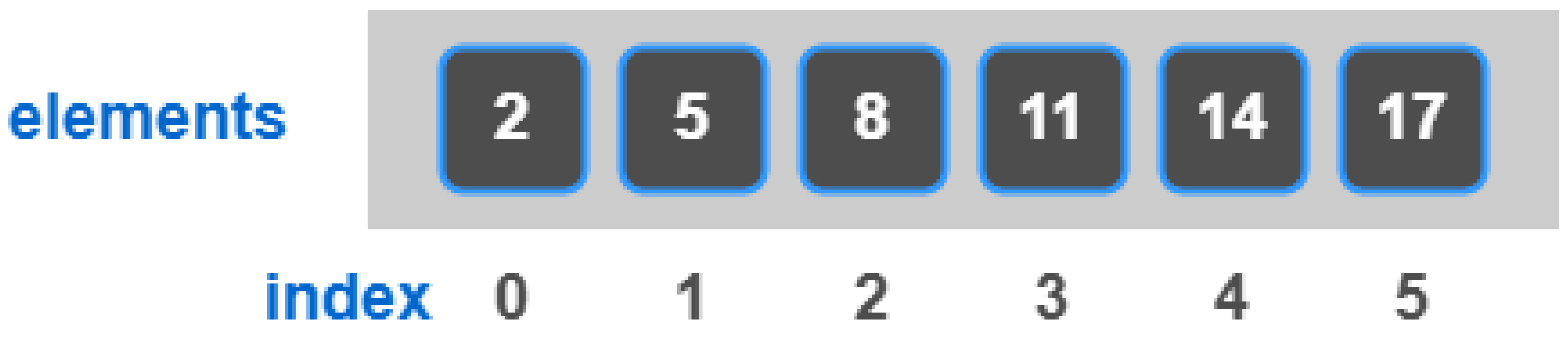
- Intro to List
- Practices
- Nested-list
- Practices

Intro to LIST

A ***list*** is a collection of items in a particular order.

In Python, square brackets (***[]***) indicate a list, and individual elements in the list are separated by commas (**,**).

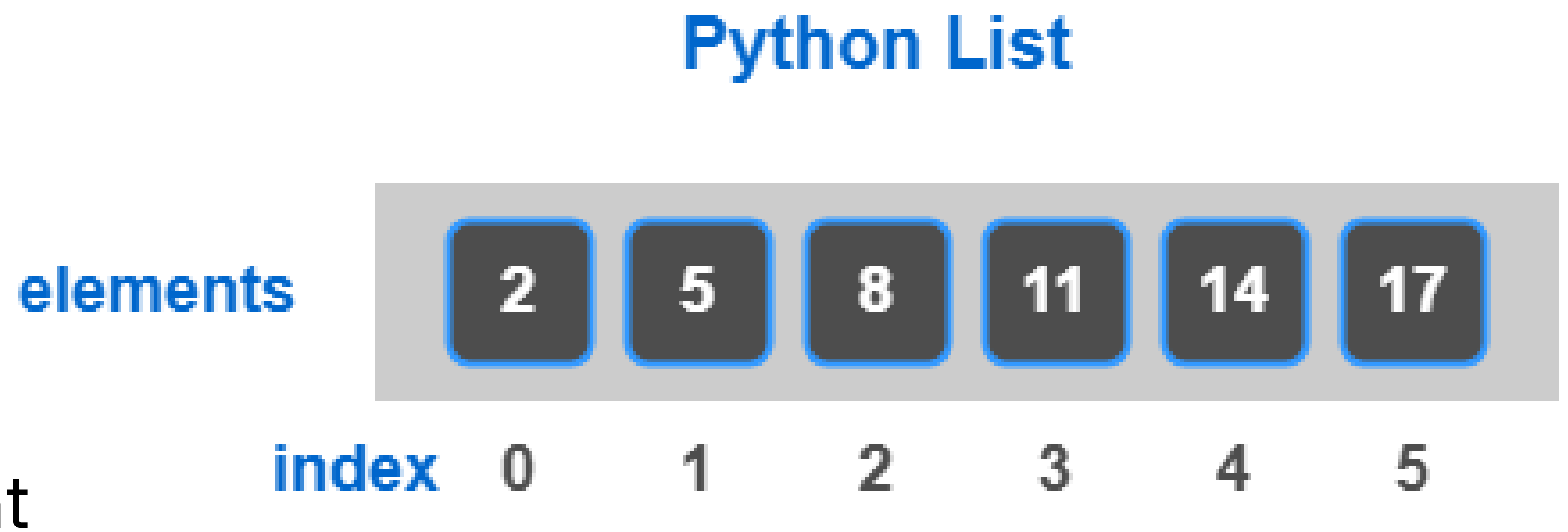
Python List



Key Components in LIST

Python ***list***:

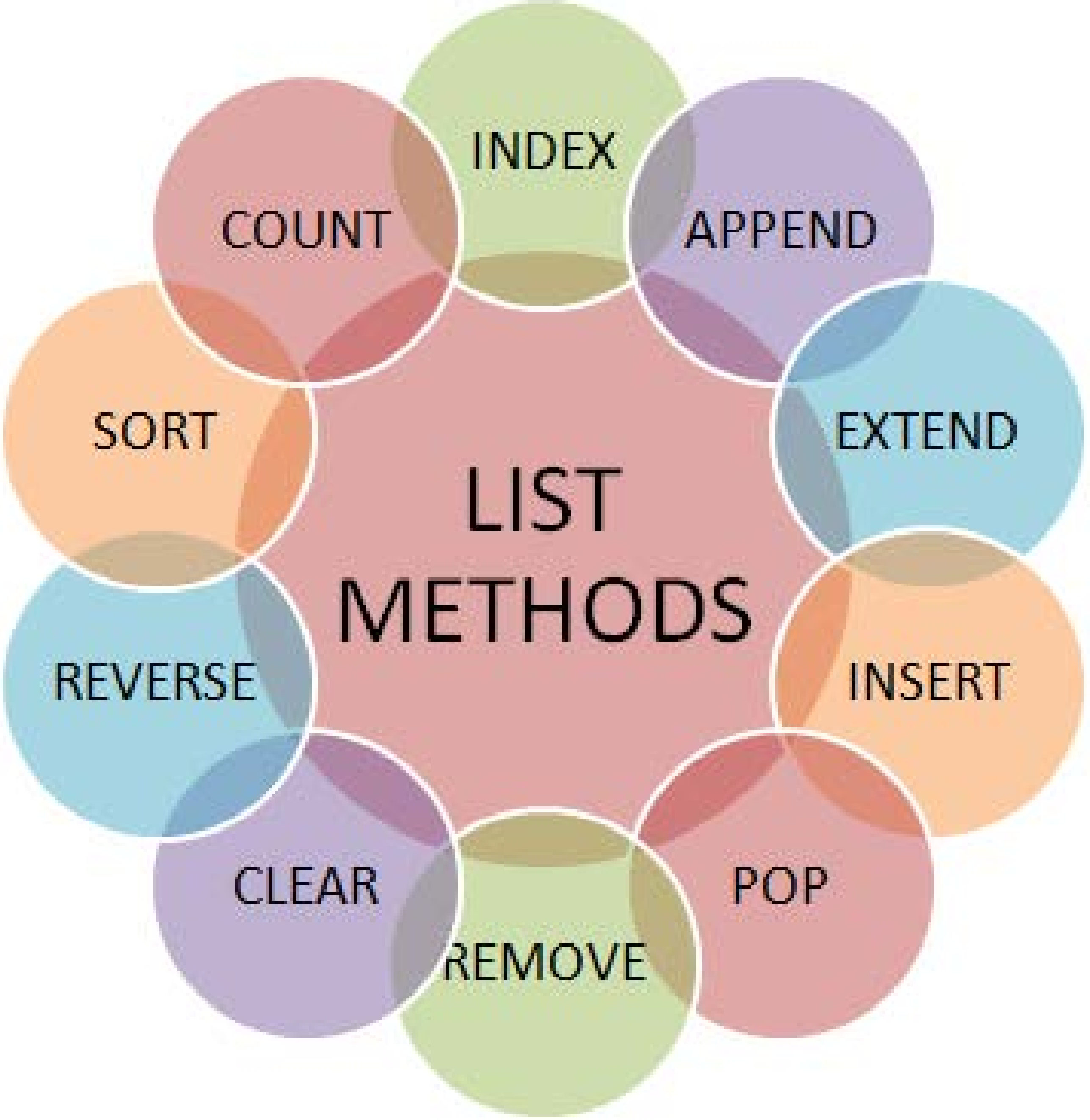
- square bracket ***[]***
- comma **,**
- each value in list is called an element
- each element is list can be accessed by its each index



Work with LIST

We can do some operation in List. For python:

1. Access - to get information from list
   - list[i]
   - count()
2. Modify - to change the element in list
   - append(), remove(), pop()
   - list[i]=value
3. Process - to work with the elements
   - sum(), max(), min()
4. Inspect - to learn about the list itselft
   - len(), copy()



Why List?

Without a List

```
student1 = "Alice"
student2 = "Bob"
student3 = "Charlie"
student4 = "David"
student5 = "Emma"
. . .
```

With a List

```
students = [
    "Alice",
    "Bob",
    "Charlie",
    "David",
. . .
]
```

Why List?

Without a List

```
student1 = "Alice"
student2 = "Bob"
student3 = "Charlie"
student4 = "David"
student5 = "Emma"
. . .
```

then, imagine if you wanna count the total or do else operation, you have to make it one by one

>> Result,
so sad if you don know list

Week4 - Condition in Python

- Intro to if-elif-else
- Practices
- Nested-if
- Practices

What?

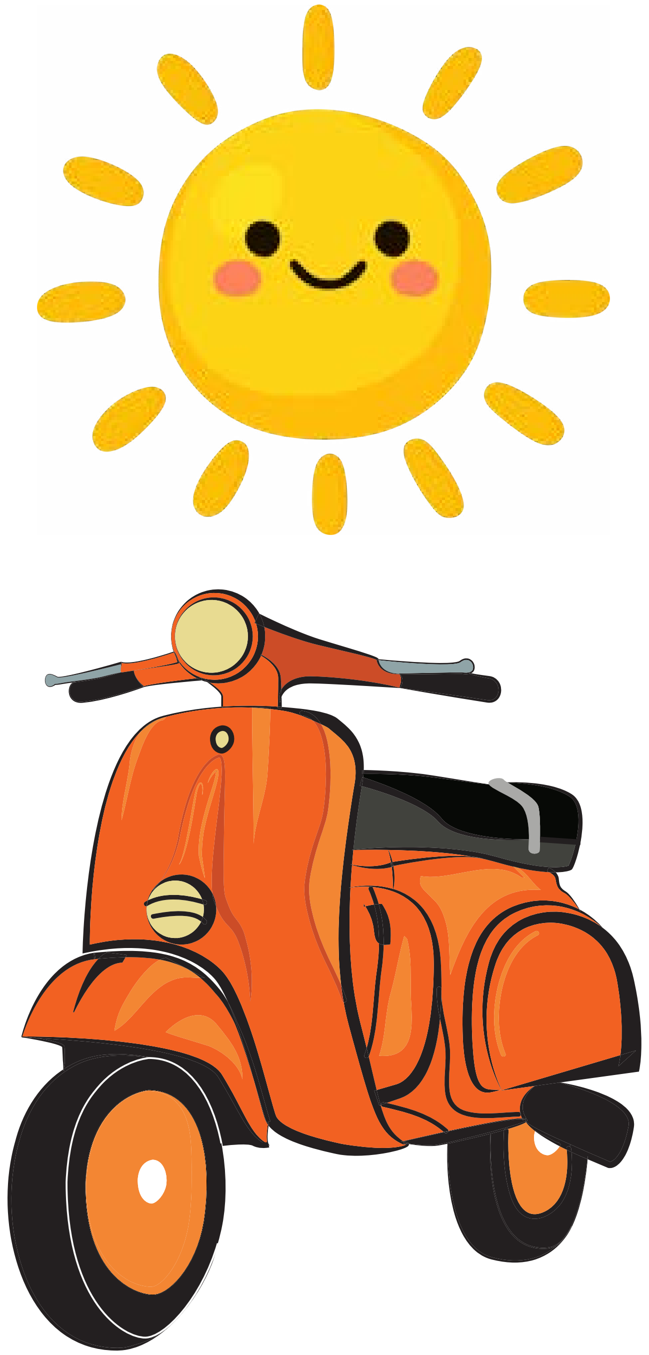
Real life, human need to make decision to inite the action.

What?

```
if weather = rain:
    booking tuk-tuk;
else:
    ride moto;
```

What?

```
if weather = rain:
    booking tuk-tuk;
else:
    ride moto;
```



Some Conditions

Python Conditions and If statements

Python supports the usual logical conditions from mathematics:

- Equals: `a == b`
- Not Equals: `a != b`
- Less than: `a < b`
- Less than or equal to: `a <= b`
- Greater than: `a > b`
- Greater than or equal to: `a >= b`

Some Logical Operators

Python Logical Operators

Logical operators are used to combine conditional statements. Python has three logical operators:

- `and` - Returns True if both statements are true
- `or` - Returns True if one of the statements is true
- `not` - Reverses the result, returns False if the result is true

Practice

Nested if

Condition needa make in number of layer of conditions.

```
Is it raining?
|
├──── YES
|     |
|     └──── Do I have enough money?
|           ├──── YES → Book tuk-tuk
|           └──── NO  → Wait
|
└──── NO
      |
      └──── Ride moto
```

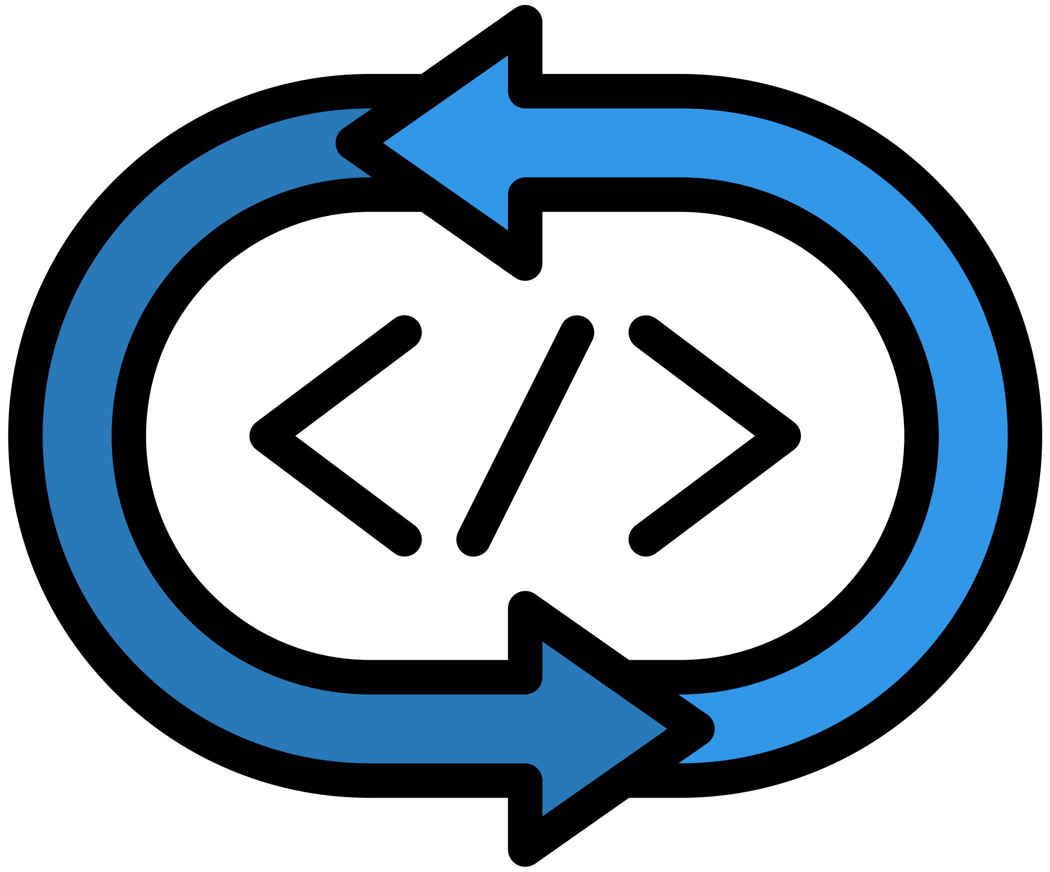
Week5 - Loop

- Intro to Loop
- Practices loop in list
- Nest-loop
- Practices loop with if-elif-else

What is Loop??

A way you need repeatedly do in a specific condition that have set.

So, a loop mean to run the same code over and over again, as long as the condition is true



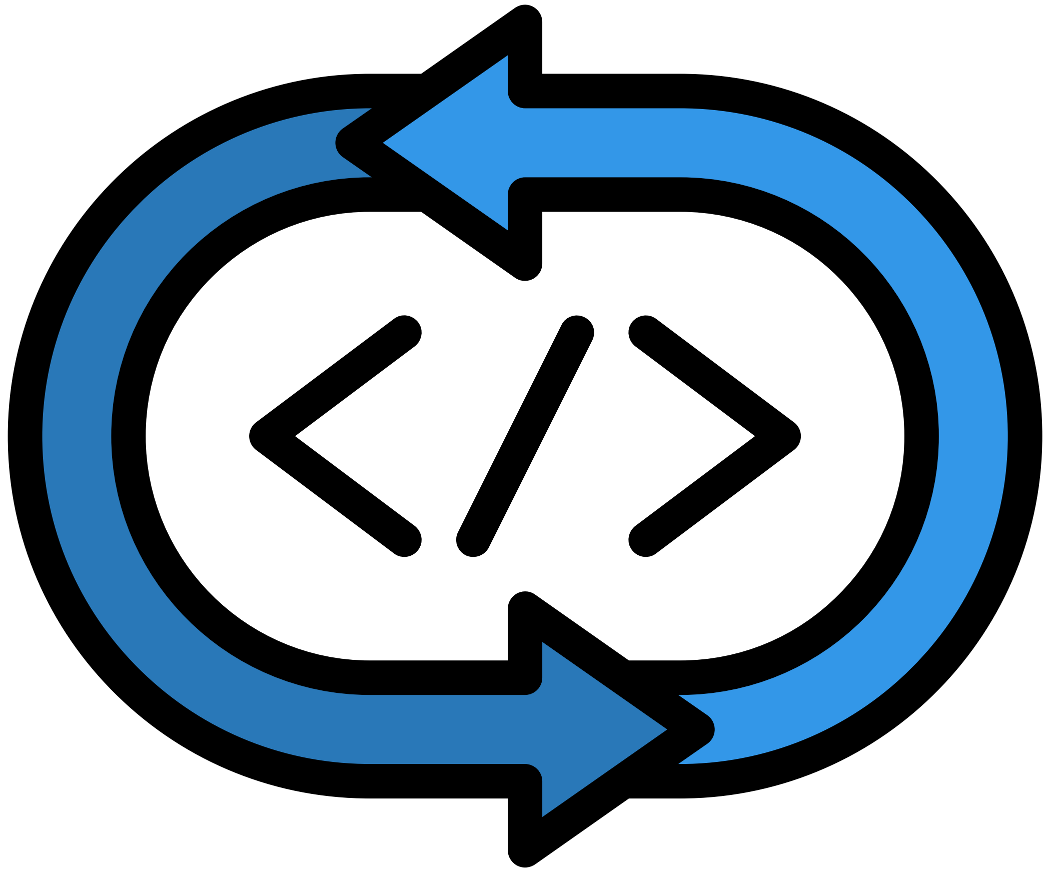
Example

The machine will always drop the items, once you pay success. Imagine if there is a line of **10, 000 people** wanna buy, **this machine keep repeat** the work.

So, this machine is being programmed with Loop method to handle this repeatative tasks.

Loop in Python

Normally, loop can be written in 2 ways: `for loop` and `while loop`



1

```
for (start, condition, step):
    code_statement;
```

2

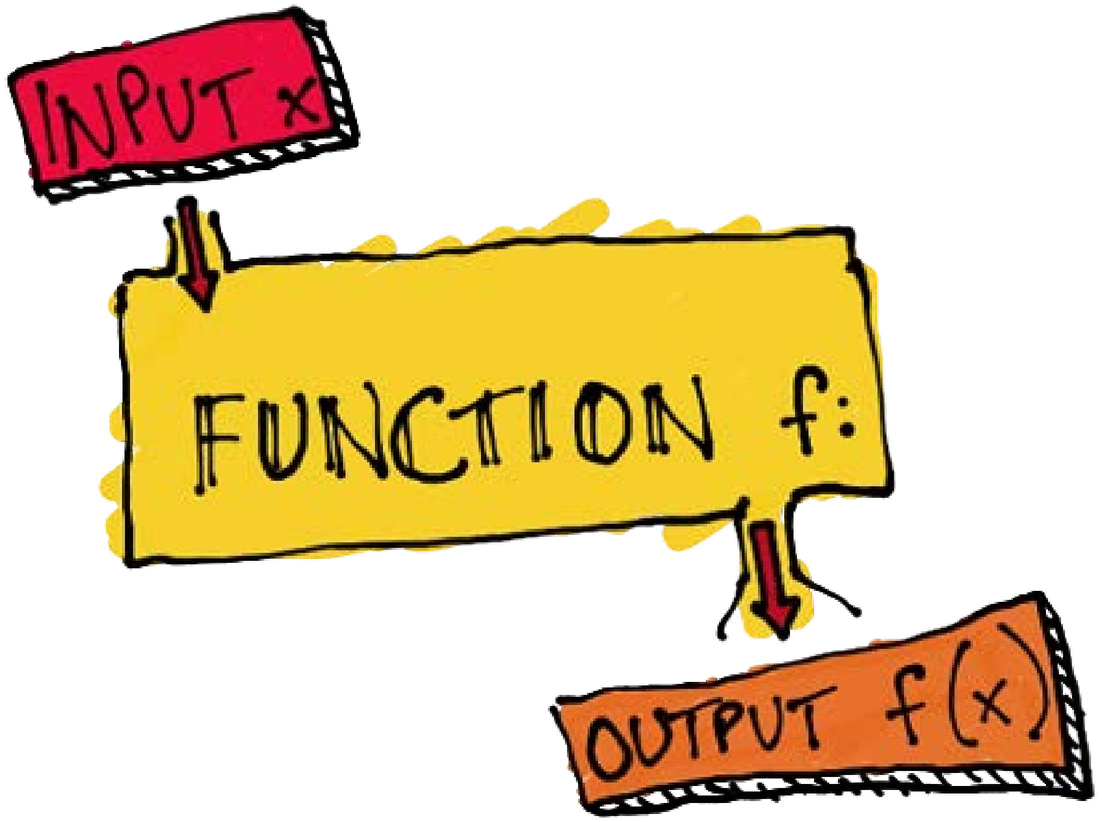
```
while condition:
    code_statement;
```

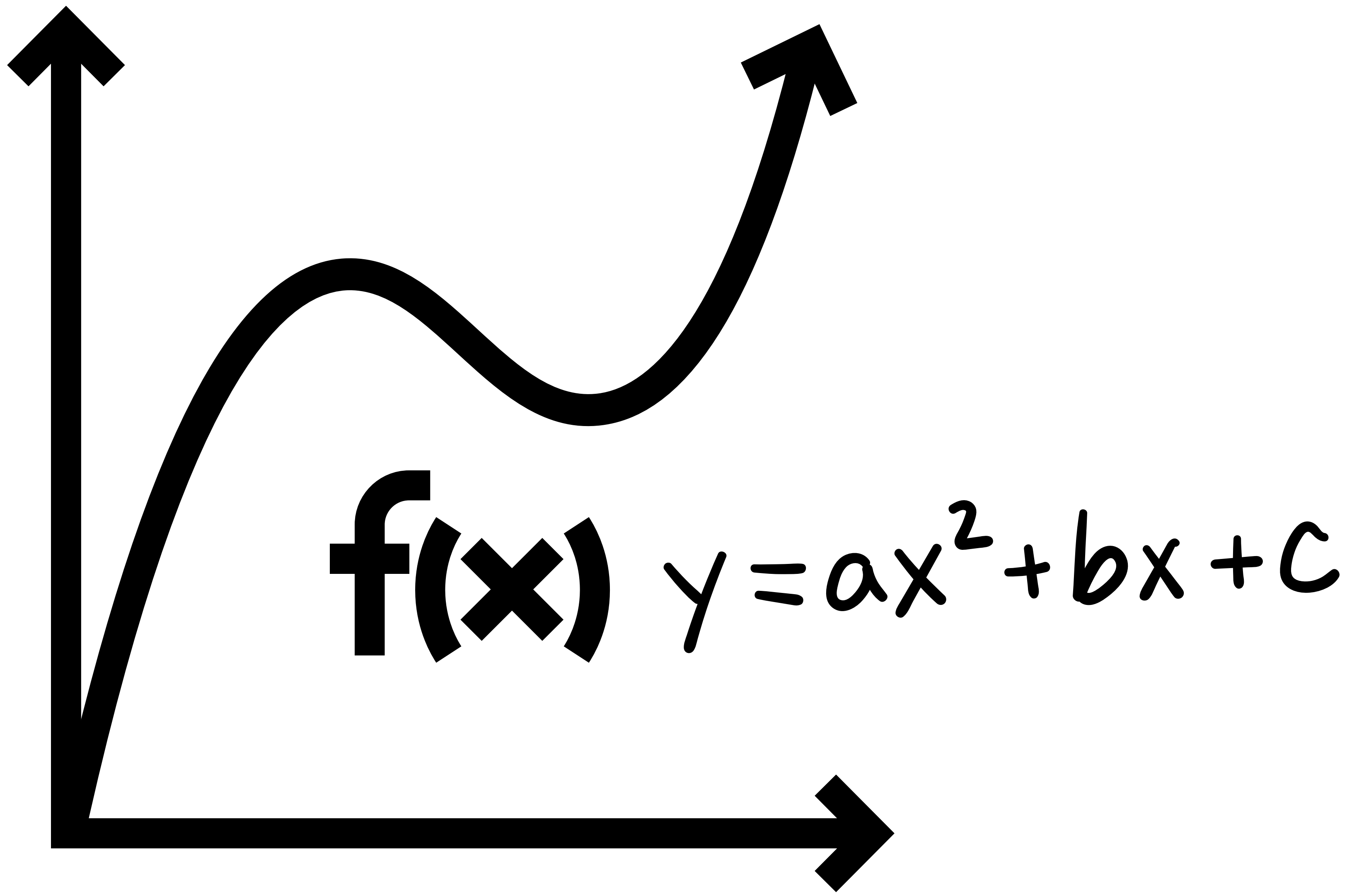
Week6 - Function

- What is function?
- Different bwt Math & Program
- Type of fucntion

What is Func??

Just like how math is working



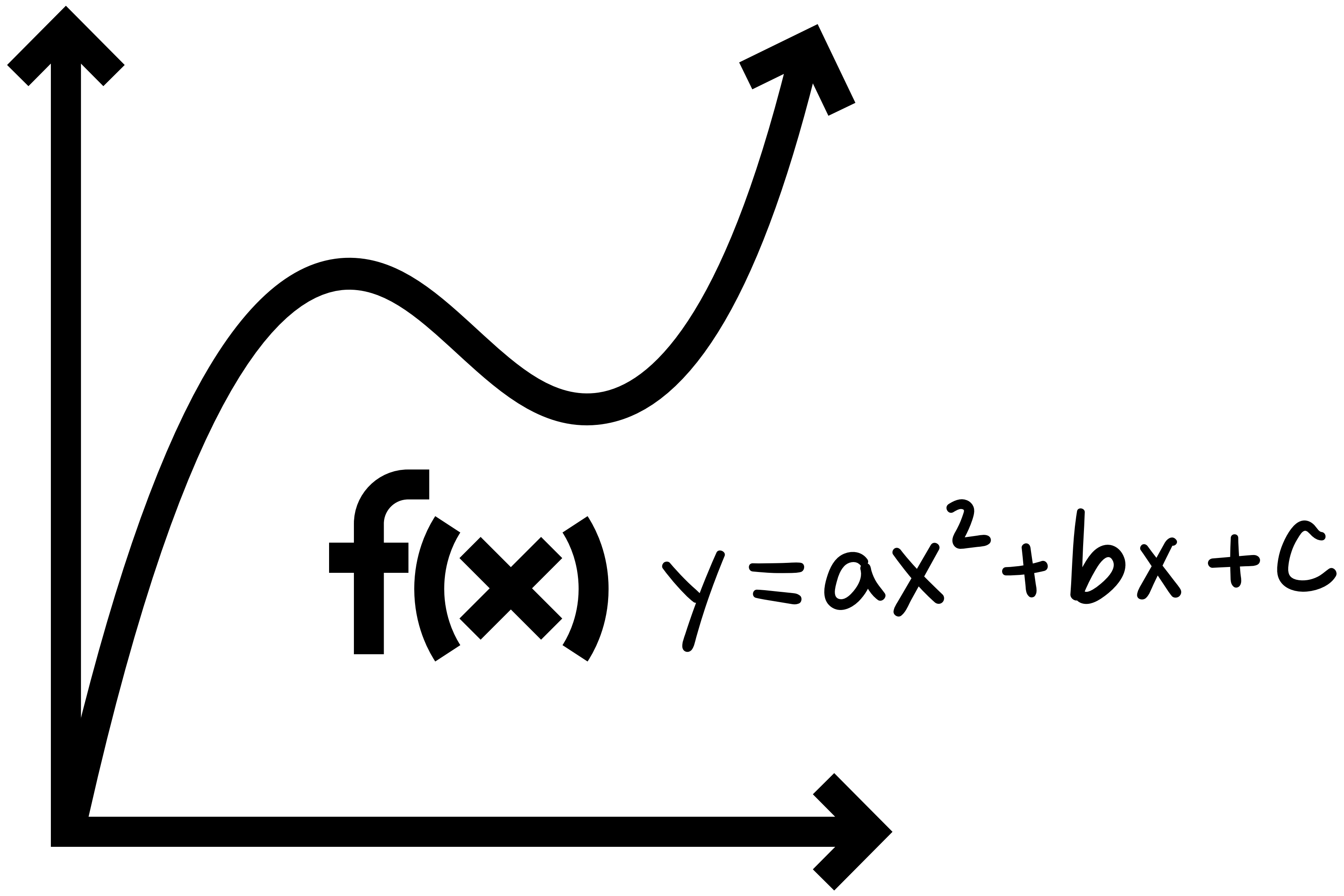


But ...

To plot the graph, we need to define how does the function of that graph is look alike based on the scenario of that exercise, right??

In mathematics, there are various ways to define based on the given scenario....

$f(x)$ $y = ax^2 + bx + c$



What is Func??

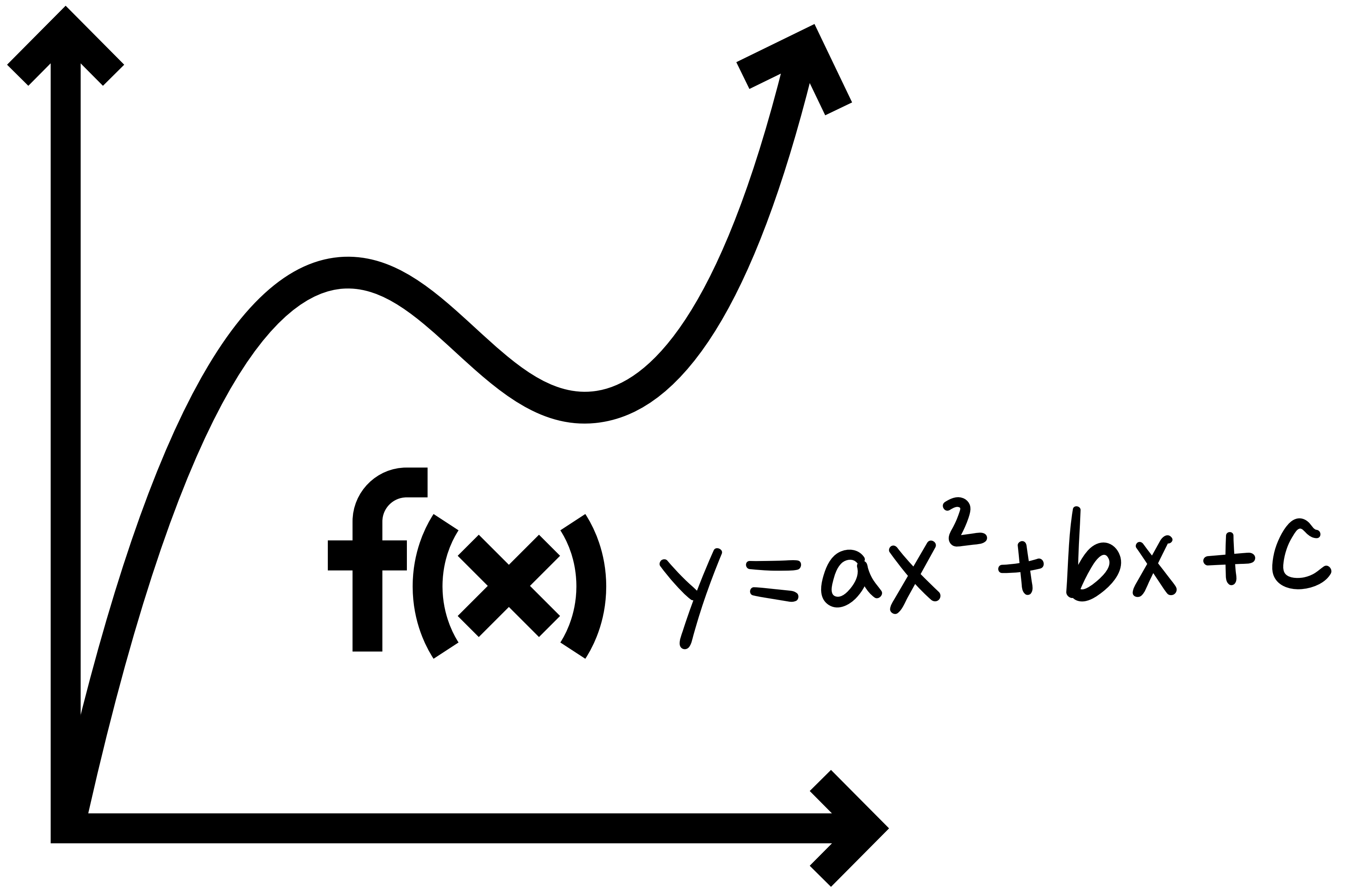
Before you plot this **f(x)** graph, what you need you have to find the **coordinator** between **(x, y)** right???

x=-1 👉 y=...

x=0 👉 y=...

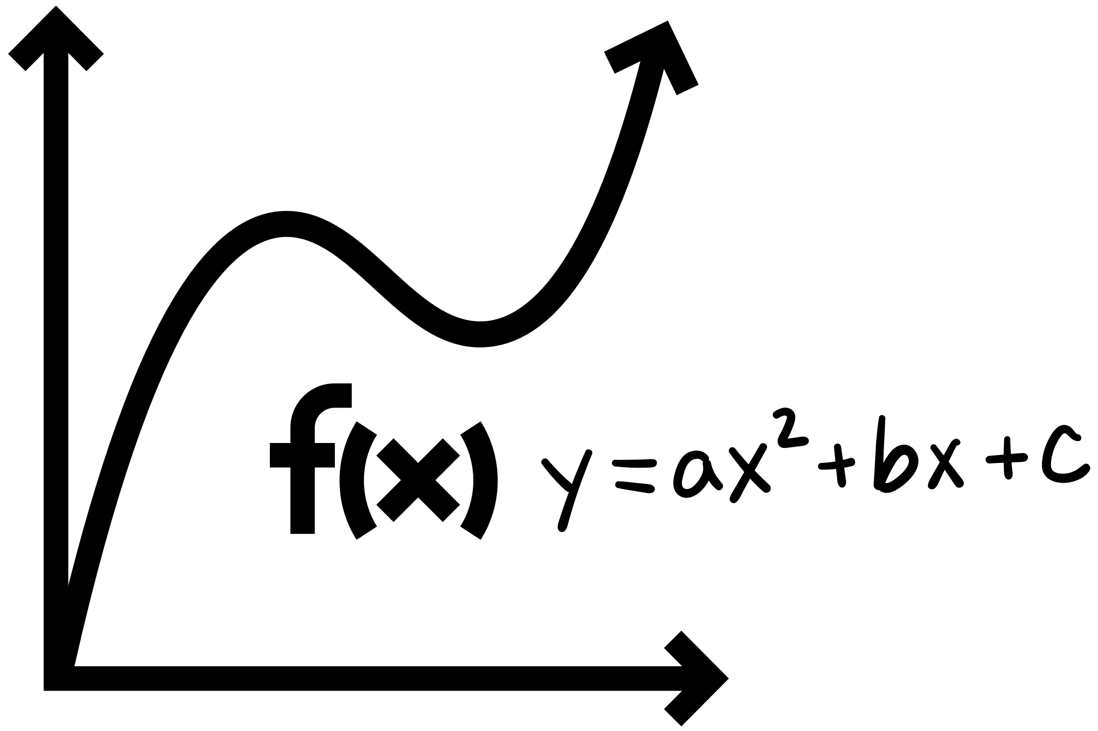
x=1 👉 y=...

...

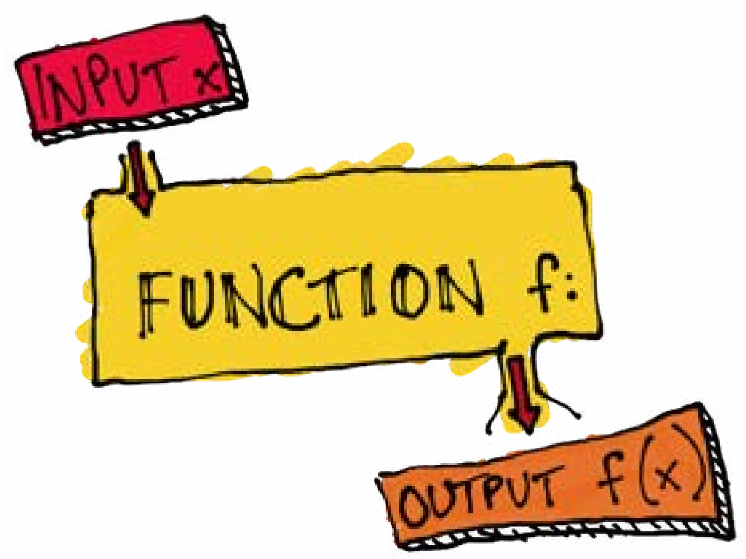


How Func is working in programming??

In mathemathics,
Different graph is defined depend on the given scenario + technique ...



In Programming,
Different algorithms is defined depend on the given scenario + language of programming ...



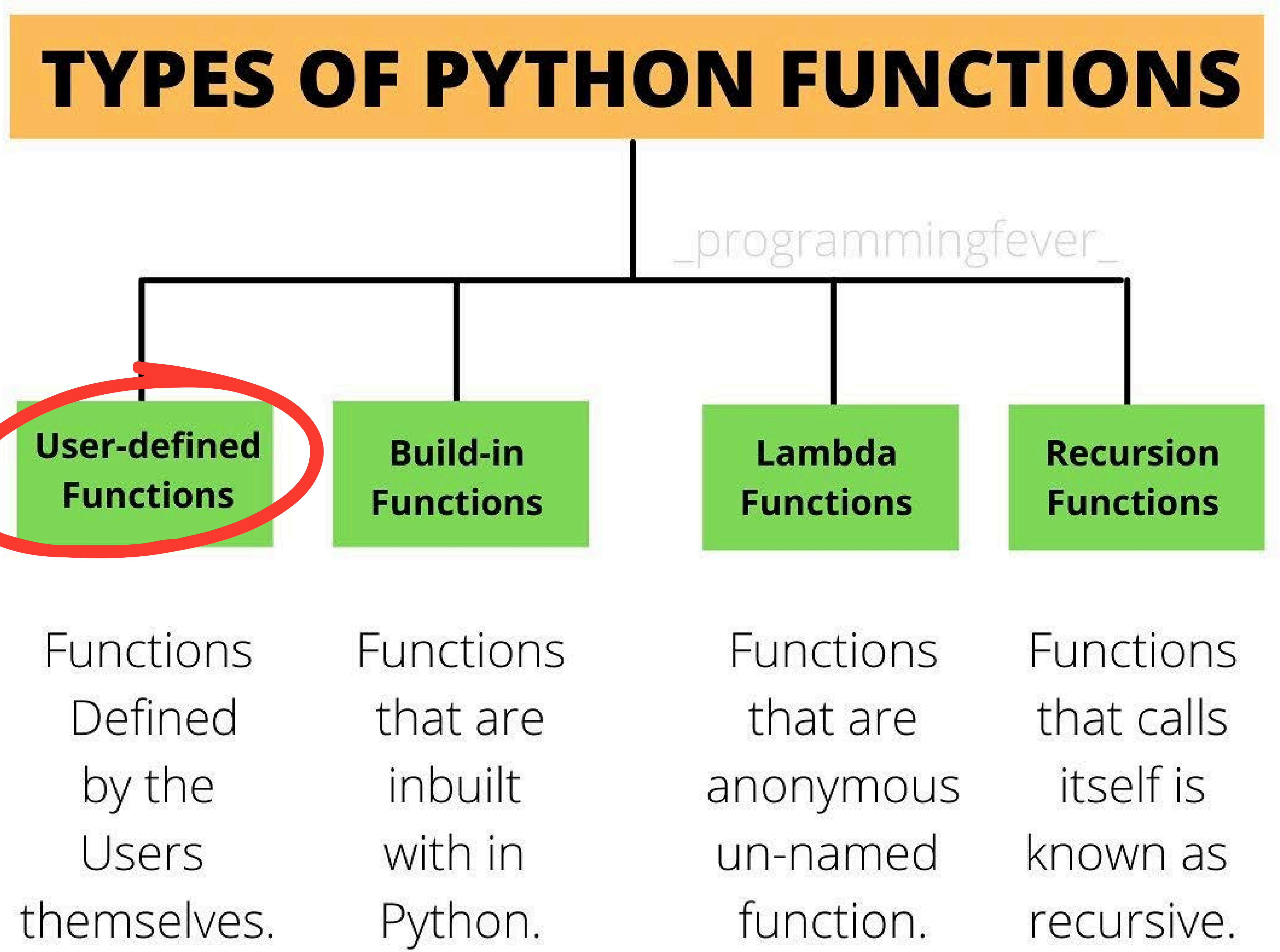
Every programming language has its each defined to write the func

TYPES OF PYTHON FUNCTIONS

programmingfever

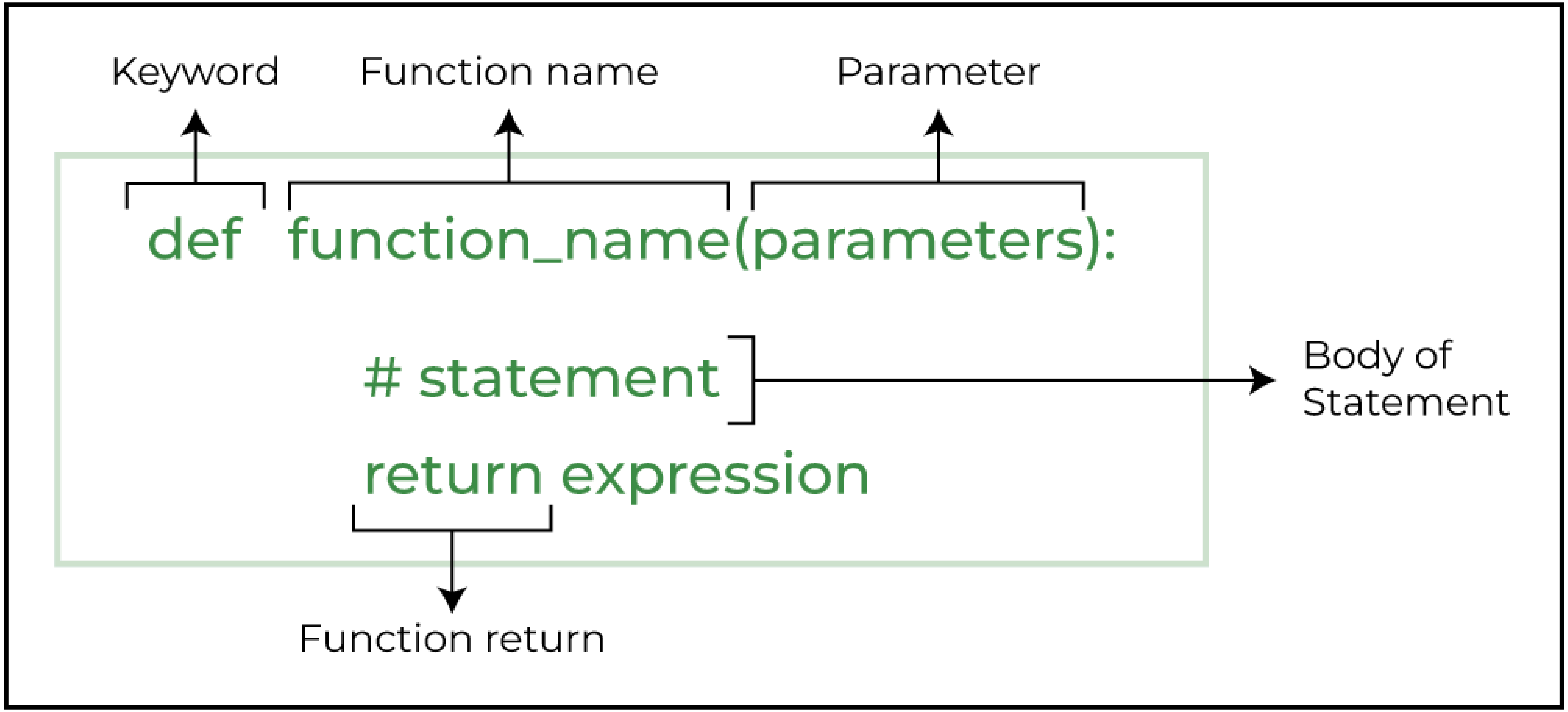
User-defined Functions	Build-in Functions	Lambda Functions	Recursion Functions
Functions Defined by the Users themselves.	Functions that are inbuilt with in Python.	Functions that are anonymous un-named function.	Functions that calls itself is known as recursive.

How to write user-defined function in python??? How many sub-type in this??



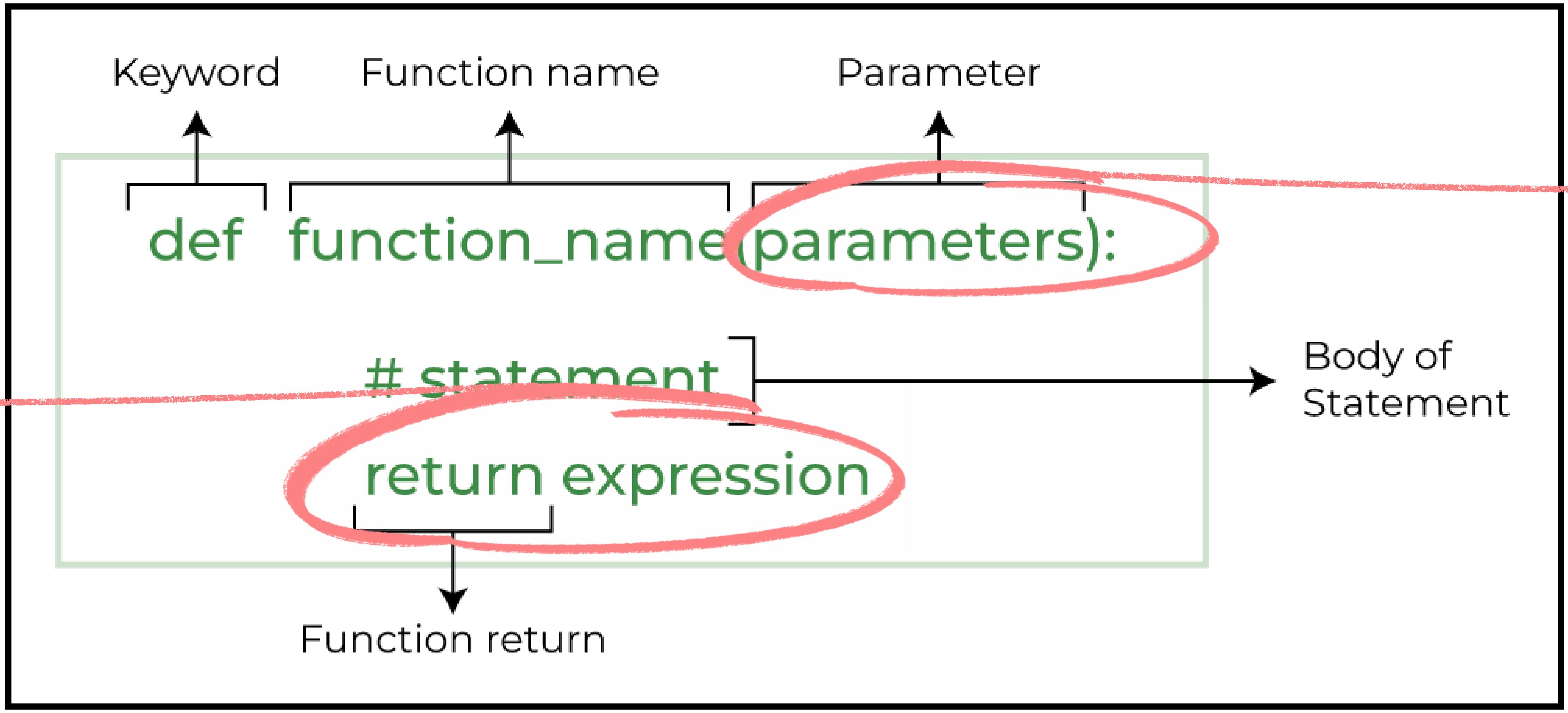
How to write the user-defined func?

Before exploring each type of user-defined function, consider to know its built-up components at first



The 4 ways, we can do the user-defined func

Before exploring each type of user-defined function, consider to know its built-up components at first



yes/no

yes/no

Pseudocode of each user-defined func:

no parameter, no return

```
def welcome():
    print("Welcome to the program")

welcome()
```

yes parameter, no return

```
def greet(name):
  print("Hello", name)

greet("Alice")
```

Pseudocode of each user-defined func:

yes parameter, no return

```
def greet(name):
    print("Hello", name)

greet("me whatever me")
```

yes parameter, yes return

```
def add_numbers(number1, number2):
  total = number1 + number2
  return total

add_numbers(5, 10)
```

There always candieees for those work hard!

@snic_n